CHARTING THE TRENDS OF GENERATIVE AI IN MATH EDUCATION: A SYSTEMATIC REVIEW

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This paper synthesizes 79 scholarly publications from recent years to map current research trends, thematic foci, and prevalent GenAI models in mathematics education. We employed a structured multi-phase protocol to guide our search, screening, coding, and analysis processes. Our findings reveal a strong focus on secondary or post-secondary levels, with the majority of studies from the U.S. and China. Notably, research has primarily centered on assessing what GenAI can do rather than investigating how it can be applied in mathematics instruction. In addition, closed-source, text-only models like ChatGPT 3.5/4 still dominate the landscape, overshadowing open-source and multimodal options. Future research directions were discussed.

Keywords: Technology; Generative Artificial Intelligence (GenAI); Math Education.

The rapid adoption of Generative Artificial Intelligence (GenAI) in education introduces both opportunities and challenges in teaching and learning math. Most existing systematic reviews on the use of GenAI in math education have focused on its potential to support teaching and learning as well as associated concerns (e.g., Almarashdi, 2024; Walkington, 2025). In this review, we adopt a global perspective by examining publication trends, key research themes, and the use of various GenAI models (e.g., open-source vs. closed-source, unimodal vs. multimodal) in math education. Three research questions guided this study.

1. What are the publication trends (grade levels, publication sources, geographical locations of the first author) in math education GenAI research?
2. What themes are studied in math education GenAI research?
3. What GenAI models have been applied in math education?

Methodology

This systematic review on GenAI in math education followed standard processes, including literature searching, data extraction, coding, analysis, and reporting (Higgins et al., 2019). An initial search retrieved 180 eligible articles from four databases: Web of Science, ScienceDirect, Google Scholar, and ERIC (via EBSCO), using the only English language, with the terms "ChatGPT," "LLM," "GenAI," "Generative AI," and "Large language model in math education OR math teaching and learning." The public type focuses on peer-reviewed journal articles, Conference proceedings, preprints, and book chapters. The others, such as posters, news articles, stories, or literature review articles, are not included. Since the release of ChatGPT in late 2022, GenAI has experienced rapid expansion in the field. Thus, all articles we identified from these databases were published between 2023 and 2024. After removing duplicates and screening for relevance, 79 articles met the preliminary inclusion criteria for the final analysis.

To explore the publication trends, the research team recorded each paper's grade level,

Zbiek, R. M., Yao, X., McCloskey, A., & Arbaugh, F. (Eds.). (2025). *Proceedings of the forty-seventh annual meeting of the North American Chapter of the International Group for the Psychology of Mathematics Education*. Pennsylvania State University. https://doi.org/ 10.51272/pmena.47.2025

geographical locations of the first author and publication resource. Then, each research member individually categorized the research themes drawing on prior studies (e.g., Martin et al., 2024). These themes included curriculum development and perspectives, attitudes, and intentions of using GenAI. We began with existing codes, but we maintained an open mind by employing a constant comparative analysis (Glaser & Strauss, 1967) to identify themes that did not align with the initial coding schemes. The final coding framework included seven categories (see Table 1), including new themes such as the use of GenAI to support mathematics teacher education and professional development. During the analysis, double coding was applied when references met the criteria for multiple categories. To explore the GenAI models applied, we analyzed the articles based on two aspects: (1) whether the GenAI model used open-source or closed-source, and (2) whether it is a single-modal or multi-modal model. To ensure the validity and triangulation of the data analysis, research members worked in pairs to code each paper. Two researchers independently coded each article, resolving discrepancies through group consensus.

Table 1: GenAI Research Themes

Themes	Definitions
Curriculum Development (Martin et al., 2024)	Designing math materials and lesson plans using GenAI tools.
Perspectives, attitudes, intentions (Martin et al, 2024)	Examining teachers' and students' attitudes, intentions, and perceptions regarding using GenAI in math education.
Teacher/Tutor	Using GenAI in mathematical teaching or tutoring.
Evaluation of GenAI ability	Assessing GenAI's capability in math (e.g., problem-solving, constructing proofs).
Student Achievement	Analyzing the effects of GenAI on students' mathematical performance.
Teacher education/professional development	Exploring the use of GenAI in supporting math teacher education/professional development.
Model Development	Creating, refining, or adapting GenAI architectures to address specific math education needs and contexts.

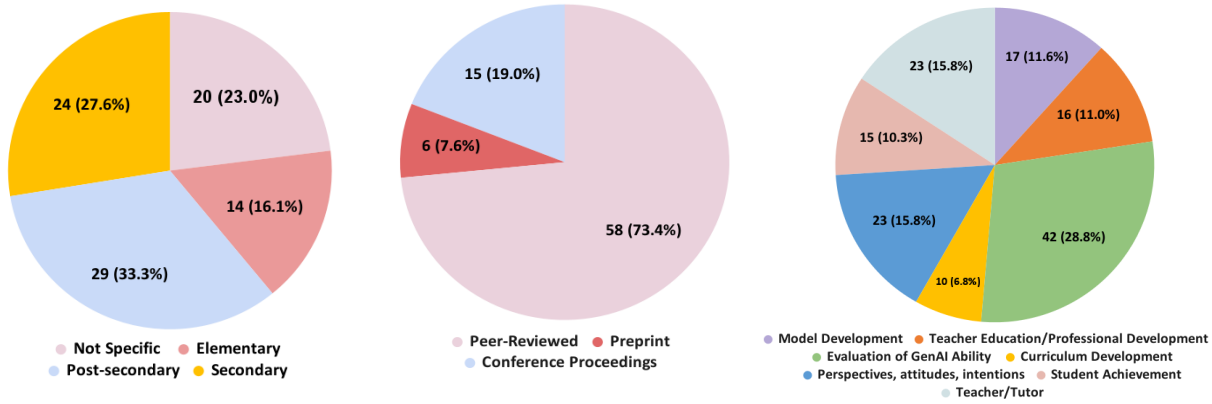
Results

Most studies on GenAI in math education have focused on secondary (24, 27.6%) and post-secondary level education (29, 33.3%), with several also addressing elementary-level education (14, 16.1%). Additionally, 20 articles did not specify the grade levels (see Figure 1a). In terms of publication sources (see Figure 1b), 58 studies (73.4%) appeared in peer-reviewed journals, 15 conference proceedings (19.0%), and six preprint papers (7.6%) published in preprint platforms such as SSRN and arXiv. The first authors represented institutions from 36 countries (see Figure 2), with the majority from the United States (21) and China (11).

Figure 1c shows that the largest group of articles is focused on the *Evaluation of GenAI Ability*, which includes 29 articles (28.8%). For example, Urhan et al. (2024) evaluated ChatGPT's capability in mathematical reasoning, especially in constructing arguments in

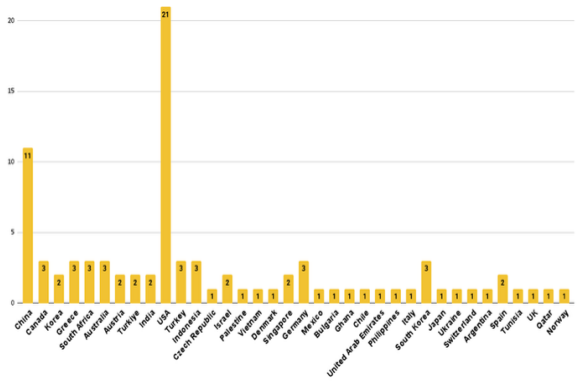
calculus. In comparison, only 10 articles (6.8%) addressed *curriculum developmen*t. For instance, Xie et al. (2024) involve high school teachers in the co-design of GenAI educational materials. The second research theme, comprising 23 articles (15.8%), has used surveys to examine students' and teachers' *perceptions* of integrating GenAI in mathematics education. The findings indicate that both groups generally hold positive attitudes toward its use in mathematics teaching and learning (e.g., Jin et al., 2024). Twenty-three articles (15.8%) have documented the use of GenAI tools to support mathematics *teaching and tutoring*. For instance, Tellez et al. (2024) developed a framework that leverages ChatGPT for formative assessment in mathematics problem-solving. Additionally, 16 studies (11%) have investigated the use of ChatGPT to support *teacher education* by assisting in the development of lesson plans and the creation of mathematical tasks in methods courses for future mathematics teachers (e.g., Noster et al., 2024). Some studies (15 studies, 10.3%) have reported that the use of GenAI positively influences students' achievements in mathematics, including problem-solving strategies and mathematical performance in elementary education (e.g., Karaman & Goksu, 2024). Lastly, 17 articles (11.6%) have investigated *Model Development,* which involves improving math problem-solving (e.g., Anh et al., 2024), virtual assistants (e.g., Pal Chowdhury et al., 2024), and automated scoring (e.g., Morris et al., 2024), among others.

Figure 1: Publication trends of GenAI in math education.



(a) Grade Level (b) Publication Sources (c) Research Themes

Figure 2: Geographical Locations of the First Author



The majority of articles (70/79, 88.6%) employed closed-source GenAI models. Among closed-source models, all employed ChatGPT families developed by OpenAI (n.d.), except one article (Matzakos et al., 2023) using the Bard model. 13 (16.4%) articles featured open-source models listed as BERT (e.g., Duan et al., 2024), LLaMA (e.g., Anh & Hung, 2024). An additional 7 (8.8%) articles employed both open-source and commercial models (e.g., Baral et al.,2023), while three articles did not specify the model type.

Most GenAI tools are single models. However, three articles (e.g., Ergene & Ergene, 2024) report the use of multimodal GenAI tools, including ChatGPT-4o, LLaVA-1.5, and DALL·E-2. In addition to these, a variety of model architectures have been examined, ranging from traditional models to advanced, large-scale generative model architectures.

Discussion and Implications

This systematic review provides an overview of the emerging landscape of GenAI research in math education, highlighting key trends and developments shaping the field. The results showed that first, geographic distributions are imbalanced. Most studies originated from the U.S. and China, consistent with broader findings (Martin et al., 2024; Lo, 2023). It may be beneficial to prioritize cross-national collaborations that leverage each region's unique strengths to innovate further and integrate GenAI into math education. While grade-level coverage is relatively balanced, existing studies emphasize older students, leaving the potential of GenAI in early childhood mathematics underexplored. Future research should explore ways to integrate AI into early childhood mathematics education (Okur et al., 2022). While 73.4% of GenAI research in math education is published in peer-reviewed journals, few articles are published in leading math education journals, with most being found in technology-focused journals. To enrich the field, greater interdisciplinary collaboration and increased submissions to core mathematics education journals are needed to bridge technological innovation with evidence-based teaching practices. Finally, the research themes indicate that the majority of studies concentrate on *Evaluating GenAI Ability* (28.8%) and studying *Perspectives, attitudes, and intentions* (15.8%), while only a small proportion (6.8%) addresses *Curriculum Development*. This disparity suggests that current efforts are predominantly centered on assessing what GenAI can do, such as its ability to solve mathematical problems or generate mathematical explanations, rather than exploring how it can be integrated into curricula and teaching practices. Moreover, our analysis revealed that only 36 articles constitute empirical studies, indicating a pressing need for more evidence-based research to guide the practical application of GenAI in mathematics education. Future research should prioritize the development and rigorous evaluation of innovative curricular models that integrate GenAI in mathematics education, alongside comprehensive assessments of its impact on teaching practices and student learning outcomes across diverse educational settings.

For model development, current research primarily relies on proprietary models (88.6%), such as ChatGPT, whose closed-source nature limits transparency, reproducibility, and customization (Huang et al., 2025). These ethical implications, such as data privacy, transparency, and potential bias in closed-source GenAI models (e.g., ChatGPT), require deeper attention. Addressing ethical considerations, including data privacy, transparency, and bias inherent in closed-source models is essential. Establishing comprehensive guidelines, clear data governance policies, professional AI ethics training, and promoting open-source alternatives (e.g., LLaMA, DeepSeek-R1) are vital to fostering trustworthy and equitable GenAI implementation (Xing et al., 2024).

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